**Expanded Triplanetary Map Notes**
**by Mike Rowles**

Given the vastness of space represented in the Triplanetary map, a larger map board might be a welcome expansion of the game.

To fit the orbits of the inner planets, the asteroid belt and Jupiter on the equivalent of two 17x22 long grain hex maps it was necessary to use a variable scale map.

Map Scale

| | |
|---|---|
| Sol to orbit of Mars, | each hex = 9.35 million km (Mkm) (0.0625 AU) |
| Mars to orbit of Ceres and Pallas | each hex = 18.7 Mkm (0.125 AU) |
| Ceres/Pallas to orbit of Saturn | each hex = 46.75 Mkm (0.3125 AU) |
| Saturn to the orbit of Pluto | each hex = 93.5 Mkm (0.625 AU) |

The orbits of Mercury, Venus, Terra, Mars, Vesta, Ceres/Pallas and Jupiter are drawn on the inner planets map. The orbits of Saturn, Uranus, Neptune and Pluto are drawn on the outer planets map.

All Planets and Asteroids orbit Sol in a counterclockwise direction. All planet movement rates assume that one game turn still equals one day. The only asteroids that move are Vesta, Ceres and Pallas. All other asteroids on the map are treated the same as the ones on the original map.

| Planet | AUs from Sun | Orbit Radius | Orbit Circumference | Movement rate |
|---|---|---|---|---|
| Mercury | 0.4 | 6 hexes | 38 | $^1/_2$ |
| Venus | 0.7 | 12 hexes | 75 | $^1/_3$ |
| Terra | 1 | 16 hexes | 100 | $^1/_4$ |
| Mars | 1.5 | 24 hexes | 150 | $^1/_5$ |
| Vesta | 2.6 | 31 hexes | 195 | $^1/_{11}$ |
| Ceres* | 2.8 | 34 hexes | 214 | $^1/_{12}$ |
| Pallas | 2.8 | 34 hexes | 214 | $^1/_{12}$ |
| Jupiter** | 5.2 | 42 hexes | 264 | $^1/_{42}$ |
| Saturn | 9.6 | 56 hexes | 352 | $^1/_{112}$ |
| Uranus | 19.2 | 71 hexes | 446 | $^1/_{159}$ |
| Neptune | 30.1 | 89 hexes | 559 | $^1/_{200}$ |
| Pluto | 39.5 | 104 hexes | 653 | $^1/_{227}$ |

Sol has a double ring of gravity hexes, the inner ring has triple gravity (3X) and the outer ring has normal gravity. Jupiter's gravity hexes have double gravity (2X).

*Ceres and Pallas are 60 degrees apart. Pallas has an inclination of 35.93° to the ecliptic but for purposes of this map it's on the same orbit as Ceres. Ceres leads Pallas.

**For most scenarios the outer planets (Jupiter and beyond) are stationary.

There are three maps and overlays for all the planets and their moons.

A 53 x 76 hex map depicting the orbits of Mercury, Venus, Mars, Terra, Vesta, Ceres, Pallas and Jupiter.

A 53 x 76 hex map depicting the orbits of Saturn, Uranus, Neptune and Pluto.

A combined 53 x 152 hex map.

A planet or asteroid's orbit should not "zig-zag". For example even though Terra's orbit on the map passes through hexes 0224, 0324 and 0223, Terra would not move through 0324. See fig 1.

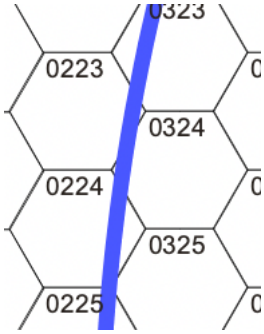


Fig. 1

To use this map it may be necessary to change the fuel capacity for all ships.

This map is unplaytested.

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